

KoylaNiti

User Manual

AI-Based Smart Governance and Compliance Monitoring System for Coal Mines
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Software Track

A complete, step-by-step reference for every role, every screen, and every workflow in the system.

All mine, workforce, contractor, and environmental data described in this manual is synthetic demo data, used to demonstrate the platform end to end.

Contents

Contents	2
1. Introduction	5
1.1 What KoylaNiti Is	5
1.2 The Core Loop	5
1.3 Synthetic Data Notice	5
2. Getting Started	6
2.1 Accessing the App	6
2.2 Demo Accounts	6
3. User Roles & Access Control	7
3.1 Ministry Admin	7
3.2 DGMS Officer	7
3.3 Mine Manager	7
3.4 Field Inspector	7
3.5 How Access Control Is Enforced	7
4. The Compliance Score	9
4.1 The Four Domains	9
4.2 How the Score Is Calculated	9
4.3 Compliance Bands	9
4.4 When Scores Recalculate	9
5. Rules — KoylaNiti Controls	10
5.1 Rule Structure	10
5.2 The Seeded Rule Set	10
5.3 Naming & Legal Citation Convention	10
6. Compliance Findings	11
6.1 What a Finding Is	11

6.2 Source Types	11
6.3 Finding Status Lifecycle	11
7. The Case Lifecycle	12
7.1 Case States	12
7.2 State Transition Rules	12
7.3 The Closure Gate	12
7.4 SLA, Due Dates & Escalation	12
8. The Compliance Evidence Graph	13
9. The Explainable Risk Panel	14
10. Government Dashboard Walkthrough	15
10.1 Home / Map	15
10.2 Mine Detail Page	15
10.3 Case Detail Page & Actions	15
10.4 Cases List & Filters	15
10.5 Contractors	15
10.6 Field Reports	16
10.7 Alerts / Notifications	16
10.8 Public Complaints (Triage)	16
11. Mine Manager Walkthrough	17
11.1 Dashboard	17
11.2 Uploading Documents	17
11.3 Submitting Remediation Evidence	17
12. Field Inspector Walkthrough	18
12.1 Assignments	18
12.2 Completing a Checklist Inspection	18
12.3 GPS & Photo Capture	18
12.4 Offline Mode & Sync	18

12.5 Filing a Standalone Field Report	18
13. AI & Document Intelligence	19
13.1 The Extraction Pipeline	19
13.2 Case Brief Generation	19
13.3 Failure Handling	19
14. Satellite Monitoring	20
15. Notifications & Escalation Logic	21
16. Contractor Management	22
17. Audit Trail	23
18. Security Model	24
19. Technology Stack	25
20. Public Complaints (Anonymous Reporting)	26
20.1 Submitting a Complaint	26
20.2 Checking Status	26
20.3 AI Summary	26
20.4 Officer Triage and Escalation	26
21. End-to-End Demo Walkthrough	27

1. Introduction

1.1 What KoylaNiti Is

KoylaNiti is a governance intelligence layer for coal mine compliance. It fuses three kinds of evidence - statutory documents, field inspections, and satellite monitoring - into one continuously-updated, explainable compliance score per mine. When a mine's score drops too low, the system automatically opens a case and routes it through a closed loop of assignment, field verification, and authorized closure, with a complete audit trail at every step.

The defensible idea behind the product is fusion plus closure: existing tools either answer legal questions (a regulation chatbot), analyze satellite imagery in isolation, or digitize a single transactional workflow. None of them combine statutory-document compliance, safety/field compliance, and environmental compliance into one continuously-scored system, and none of them close the loop from a detected problem to a verified, audited resolution. KoylaNiti's whole design is built around doing that one thing well, rather than building thirteen disconnected feature screens.

1.2 The Core Loop

Every feature in the system exists to serve one repeating cycle: **Obligation** → **Evidence** → **Risk** → **Action** → **Verification** → **Audit**. A regulatory obligation (a KoylaNiti Control, grounded in a real statute or DGMS circular) is checked against evidence - an uploaded document, a field inspection, or a satellite observation. Where evidence fails a check, a compliance finding is raised and feeds the mine's risk score. If the score crosses a threshold, a case is opened and assigned to a Field Inspector. The inspector's field evidence is submitted, a DGMS Officer verifies it, and the case is formally closed - with every step permanently recorded in an audit timeline.

1.3 Synthetic Data Notice

This is a hackathon prototype. Every mine, user, document, contractor, and finding described in this manual is synthetic demo data, generated to make the workflow demonstrable end to end. Internal rule identifiers (e.g. CMR-SAFE-001) are labelled throughout the app as "KoylaNiti Control ID"s - they are the system's own internal references, not official regulation citations, even where a rule's description points at a real statute (such as CMR 2017 Regulation 106(2)/(3)) for realism. AI in this system assists document field-extraction and writes plain-language case summaries; it never decides legal compliance - that judgment always stays with a human DGMS Officer.

2. Getting Started

2.1 Accessing the App

Open the app in a browser (locally at `http://localhost:3000`, or the deployed URL if hosted). You will land on the login page. There is a single login form for all roles - the system determines your role and redirects you automatically after signing in:

- **Ministry Admin** and **DGMS Officer** land on `/gov`
- **Mine Manager** lands on `/manager`
- **Field Inspector** lands on `/inspector`

If you try to open a URL that belongs to a different role than your account, you are redirected back to your own role's home page automatically - this is enforced both at the page-routing level and, more importantly, independently re-checked by the backend on every single API request.

2.2 Demo Accounts

Every demo account uses the same password: **demo-2026**.

Username	Role	Scope
admin	Ministry Admin	All mines, all states
dgms_east	DGMS Officer	Chhattisgarh only (covers the hero mine, MINE-1003)
dgms_south	DGMS Officer	Telangana only
manager1 ... manager8	Mine Manager	One per seeded mine - manager3 owns the hero mine
inspector1	Field Inspector	Ramesh Patil - own assignments only
inspector2	Field Inspector	Sunita Rao - own assignments only
inspector3	Field Inspector	Arjun Nair - own assignments only

3. User Roles & Access Control

The system has exactly four roles. Each one has a distinct home area, a distinct set of actions it can take, and a distinct data scope - and that scope is enforced on the server for every request, not just hidden in the interface.

3.1 Ministry Admin

The top-level oversight role.

- Sees every mine, in every state, with no restriction.
- Can view and act on any case, exactly like a DGMS Officer, anywhere in the country.
- Sees the full contractor list, all field reports, and ministry-wide alerts.

3.2 DGMS Officer

The regional regulator - the role that actually drives cases to closure.

- Scoped to a single state (their `jurisdiction_state`) - a DGMS Officer for Chhattisgarh cannot see mines, cases, or findings in Telangana, and the reverse.
- Reviews a mine's score breakdown and open findings.
- Assigns a case to a Field Inspector, optionally setting a due date (SLA) and an escalation target.
- Verifies submitted field evidence - this is the step that resolves the linked findings.
- Formally closes (resolves) a case once verified, recording a closure reason.
- Reviews standalone field reports and marks them reviewed.

3.3 Mine Manager

Represents one specific mine - the operator's side of the platform.

- Scoped to exactly one mine (set on their account at seed time).
- Sees their own mine's score breakdown, documents, open findings, and open cases.
- Uploads compliance documents (clearances, permits, certificates) with the document's key text, which the AI pipeline reads to extract structured fields.
- Submits remediation evidence on a case once corrective action has been taken on site.

3.4 Field Inspector

The boots-on-the-ground role - the one that checks reality against the record.

- Sees only the cases assigned specifically to them.
- Has unrestricted read-only visibility of all mines (needed to file a standalone report on a mine that has no active assignment) - but can still only submit inspections and see cases that are actually theirs.
- Fills out a pass/fail safety checklist on site, captures a real GPS location and a photo, and submits it as evidence.
- Can save an inspection locally while simulating offline conditions, and sync it later in one batch.
- Can file a standalone field report (an incident, an environmental observation, an operational exception) that is not tied to any existing case assignment.

3.5 How Access Control Is Enforced

Every account carries a signed JWT access token, issued at login and attached to every API request as an Authorization: Bearer header. Two layers enforce access:

- **Edge routing (optimistic):** the web app checks that a valid, unexpired token exists and redirects you toward your own role's area if you try to open a route belonging to another role. This is a convenience check only - it does not query the database.
- **Backend enforcement (authoritative):** every API endpoint re-derives your role and scope from the token against the database on every single call, and filters every query accordingly - by jurisdiction state for a DGMS Officer, by mine ownership for a Mine Manager, by assignment for a Field Inspector. A Field Inspector requesting a case that was never assigned to them receives a 404, not the record.

4. The Compliance Score

4.1 The Four Domains

Every mine's score is built from four weighted domains, never a single opaque number:

Domain	Weight	What it measures
Statutory Documents	30%	Are clearances, permits, and certificates valid and not expired?
Safety / Field Inspection	35%	Do on-site checklist results show real safety compliance?
Environmental / Satellite	25%	Any unexplained vegetation loss or lease-boundary encroachment?
Operational / Other	10%	Other configured indicators, e.g. statutory returns filed on time.

4.2 How the Score Is Calculated

Each domain starts at 100. Every currently OPEN compliance finding in that domain subtracts a fixed number of points based on its severity (CRITICAL 40, HIGH 25, MEDIUM 15, LOW 5), and the domain score is clamped to the 0-100 range. The mine's overall score is the weighted sum of the four domain scores. Because the calculation is a plain, fixed formula over real findings - never an AI's own judgment - the score can always be explained by pointing at the specific findings behind it.

Example: a mine with one CRITICAL statutory finding (Statutory domain drops to 60), one HIGH and one CRITICAL safety finding (Safety domain drops to 35), one CRITICAL environmental finding (Environmental domain drops to 60), and a clean Operational domain, scores: $60 \times 0.30 + 35 \times 0.35 + 60 \times 0.25 + 100 \times 0.10 = 55.25$.

4.3 Compliance Bands

GREEN	80 - 100 — Compliant, no action needed.
YELLOW	60 - 79 — Some issues, worth watching.
RED	Below 60 — A case is opened automatically.

4.4 When Scores Recalculate

The score is recomputed live every time a mine's compliance page is opened, and immediately after any event that changes a finding's status: a document's AI extraction completing, a DGMS Officer verifying a case (which resolves its linked findings), or a case being closed. If a recalculation causes a mine to newly cross into the RED band and it has no other open case, the system automatically creates one, linking every currently open finding on that mine to it.

5. Rules — KoylaNiti Controls

5.1 Rule Structure

A rule (displayed in the app as a "KoylaNiti Control ID") is a versioned record, not a hardcoded check buried in application code - this is deliberate, so the rule set can grow without changing software. Each rule carries: a stable ID, a domain, a source citation, a plain-language description, a check type (DATE / BOOLEAN / NUMERIC / GEO / INSPECTION), a threshold configuration, a severity, and a description of what evidence satisfies it.

5.2 The Seeded Rule Set

Control ID	Domain	Severity	Checks
CMR-STAT-001	Statutory	CRITICAL	Environmental Clearance not expired
CMR-STAT-002	Statutory	HIGH	Forest Clearance not expired
CMR-STAT-003	Statutory	HIGH	Mining Plan / Scheme approval current
CMR-STAT-004	Statutory	MEDIUM	Consent to Operate not expired
CMR-SAFE-001	Safety	HIGH	Statutory safety inspection not overdue (CMR 2017 Reg. 106(2)/(3))
CMR-SAFE-002	Safety	CRITICAL	Ventilation / gas monitoring checklist item passes
CMR-SAFE-003	Safety	MEDIUM	Fire-fighting / emergency equipment available
CMR-ENV-001	Environmental	CRITICAL	No vegetation loss beyond the approved lease boundary
CMR-ENV-002	Environmental	HIGH	Progressive reclamation at least 70% of plan
CMR-OPS-001	Operational	LOW	Production / statutory returns filed on time

5.3 Naming & Legal Citation Convention

Wherever a rule ID appears in the app, it is always shown as "KoylaNiti Control ID: <id>" - never as a bare code - specifically so it can never be mistaken for an official government regulation number. The hero control behind the demo's central case, CMR-SAFE-001, cites CMR 2017 Regulation 106(2)/(3) as its source, but its description is written to avoid asserting a specific inspection cadence (e.g. "monthly") as though quoted from the regulation itself - the 90-day threshold used to flag an overdue inspection is this system's own configurable check parameter, not a claim about the statute's literal wording.

6. Compliance Findings

6.1 What a Finding Is

A finding is a single, concrete failure of a rule at a specific mine - e.g. "Environmental Clearance expired 2026-07-27" or "Ventilation/gas monitoring checklist item failed at last recorded inspection." Every finding records which mine it belongs to, which rule (if any) it violates, its severity, a plain-language description, and how many score points it currently costs that mine.

6.2 Source Types

- **DOCUMENT** — raised automatically when an uploaded document's AI-extracted expiry date has already passed.
- **INSPECTION** — raised from field checklist results (seeded in the demo dataset; the live checklist submission flow records answers but does not yet auto-generate new findings from a failed checklist item in this build).
- **SATELLITE** — raised from the mocked satellite monitoring pipeline (see section 14).
- **OPERATIONAL** — other configured indicators, e.g. a late statutory return.

6.3 Finding Status Lifecycle

- **OPEN** — actively counted against the mine's score.
- **RESOLVED** — no longer counted. A finding becomes RESOLVED specifically when a DGMS Officer verifies the case it is linked to (see section 7.3) - never automatically, and never just because a case was closed.
- **ACKNOWLEDGED** — a status the data model supports for future use; not currently set by any workflow in this build.

7. The Case Lifecycle

7.1 Case States

- DETECTED
- TRIAGED
- ASSIGNED
- INSPECTION / REMEDIATION
- EVIDENCE SUBMITTED (shown as "VERIFICATION PENDING")
- VERIFIED
- CLOSED

7.2 State Transition Rules

Transitions are only ever allowed in one direction along this exact chain, and every single transition is checked and recorded on the server - there is no way to skip a step from the interface or by calling the API directly. Each transition also writes a permanent entry to the case's audit timeline (see section 8), including who performed it and when.

From	To	Triggered by
DETECTED / TRIAGED	ASSIGNED	A DGMS Officer assigns the case to a Field Inspector
ASSIGNED / INSPECTION-REMEDIATION	EVIDENCE SUBMITTED	An inspection is submitted against the case, or the Mine Manager submits remediation evidence
EVIDENCE SUBMITTED	VERIFIED	A DGMS Officer verifies the submitted evidence
VERIFIED	CLOSED	A DGMS Officer resolves the case with a closure reason

7.3 The Closure Gate

This is a deliberate integrity rule, not an incidental detail: **a case cannot be closed while any finding linked to it is still OPEN**. Verifying a case (EVIDENCE SUBMITTED → VERIFIED) is the specific moment a DGMS Officer confirms the submitted evidence actually addresses the findings, and that is when those findings become RESOLVED. If someone attempts to close (resolve) a case while a linked finding is still open - for example, if a finding was reopened after verification - the API rejects the request outright rather than allowing a case to show as CLOSED with unresolved problems behind it.

7.4 SLA, Due Dates & Escalation

When assigning a case, a DGMS Officer may optionally set a due date (the SLA) and a free-text escalation target (e.g. "State DGMS Office") describing who the case escalates to if it slips. A case is flagged OVERDUE automatically once its due date has passed and it is not yet CLOSED - this is computed live on every read, not stored as a separate flag that could go stale. If a case is assigned with a due date that is already past or within 48 hours, the system immediately raises an in-app alert to the relevant oversight officers (see section 15).

8. The Compliance Evidence Graph

This is the platform's central differentiator: every case detail page shows the full chain connecting a regulatory obligation to its eventual, verified resolution - not just a score or an isolated case record. Each node is populated from real underlying data and is empty/honest ("Not verified yet", "No field inspection submitted yet") wherever a step hasn't happened yet, rather than faking a filled state.

#	Node	Populated from
1	Regulation / KoylaNiti Control	The rule(s) behind the case's linked findings, shown with full source citation and description
2	Evidence / Document	Any DOCUMENT-sourced findings linked to the case
3	Field Observation	Inspections submitted against the case (checklist results, GPS, photo)
4	Risk Score Contribution	How many points the case's findings are costing the mine's score, and the case's own severity
5	Corrective Action	The case itself - title, current status, and who it's assigned to
6	Closure Evidence	The audit entry recorded when the case was closed, including the stated reason
7	Authorized Verification	The audit entry recorded when a DGMS Officer verified the case, including who verified it and when
8	Audit Timeline	Every recorded event on the case, in chronological order, from automatic detection through to closure

9. The Explainable Risk Panel

Both the mine detail page and the case detail page show a "Why is this high risk?" panel, which turns the current open findings into a plain-language list of factors rather than leaving a judge, official, or manager to infer risk from a number alone. It leads with an aggregate line (e.g. "3 CRITICAL findings currently open") and then lists each contributing finding with its severity.

The panel always carries two permanent disclaimer lines, by design: *"Prototype risk assumptions - demonstration only"* and *"AI/ML assists prioritisation; it does not decide legal compliance."* This is a direct, deliberate guard against the score or its factors ever being presented as an automated legal determination.

10. Government Dashboard Walkthrough

This is the area used by Ministry Admin and DGMS Officer accounts.

10.1 Home / Map

- Stat tiles: total mines, count of RED mines, count of YELLOW mines, and average score - all scoped to what the logged-in role can see.
- An interactive map (Leaflet/OpenStreetMap) plotting every visible mine as a colour-coded pin - green, yellow, or red matching its current band. Clicking a pin opens a popup with the mine's name, score, and a link into its detail page.
- A "Cases needing action" list - every case currently at DETECTED, TRIAGED, or ASSIGNED, newest first.
- A full list of every visible mine with its current score and band.

10.2 Mine Detail Page

Opened by clicking any mine (from the map, a case, or the mine list). Shows, top to bottom:

- Header: mine name, location, current score and band, and a link into its open case if one exists.
- The explainable risk panel (section 9).
- Score breakdown: a progress bar per domain, each labelled with its weight.
- The full list of open findings, each with its KoylaNiti Control ID, domain, source type, and severity.
- Documents on file, with extraction status and expiry date where known.
- A satellite before/after comparison, if the mocked satellite pipeline has data for that mine (section 14).

10.3 Case Detail Page & Actions

Opened from a case list or from a mine's open-case link. Shows the risk panel, the full Evidence Graph (section 8), the list of findings linked specifically to this case, any inspection evidence submitted against it (checklist results, GPS, photo), and an actions panel that changes based on the case's current status:

- **DETECTED / TRIAGED:** a form to assign the case to a Field Inspector, with optional due date and escalation target fields.
- **ASSIGNED / INSPECTION-REMEDIATION:** a waiting message showing the due date and escalation target if set.
- **EVIDENCE SUBMITTED:** a "Verify evidence" button.
- **VERIFIED:** a closure-reason text box and a "Close case" button.
- **CLOSED:** the closure timestamp.

10.4 Cases List & Filters

A full list of every case visible to the logged-in account, filterable by status via quick-filter pills across the top. Each row shows the case title, mine, case ID, age, severity, an OVERDUE badge where applicable, and current status.

10.5 Contractors

A list of registered contractors, each with a specialization and license number. Opening one shows their profile, a count of related findings and open actions, and the specific findings linked to them - each clicking through to its mine or case. This is intentionally a lightweight governance relationship view, not a full contractor-management ERP.

10.6 Field Reports

A triage list of standalone field reports - inspections a Field Inspector filed that were not tied to an existing case assignment (section 12.5). Each entry shows the report type, mine, inspector, timestamp, and notes, with a "Mark reviewed" action that logs the review to the audit trail.

10.7 Alerts / Notifications

A dedicated Alerts page (also reachable via the bell/unread-count indicator in the header) listing every in-app notification for the logged-in account, each with a "Mark read" action. See section 15 for what triggers a notification.

10.8 Public Complaints (Triage)

A triage queue for anonymously submitted public complaints (see section 20 for the full public-facing flow). Each entry shows category, mine, an AI-generated summary, and status; an officer can dismiss a complaint with notes or escalate it into a real case, which then follows the ordinary case lifecycle (section 7) unchanged.

11. Mine Manager Walkthrough

11.1 Dashboard

Shows the manager's own mine only: current score and band, the score breakdown, open cases at that mine, and the documents on file with their extraction status.

11.2 Uploading Documents

From the Documents page, a manager selects a document type (Environmental Clearance, Forest Clearance, Mining Plan Approval, Consent to Operate, Safety Certificate, or Other), pastes or types in the document's key text (standing in for OCR in this prototype), and optionally attaches the file itself. Submitting it kicks off the AI extraction pipeline in the background (section 13) - the page does not wait for it to finish.

11.3 Submitting Remediation Evidence

On an assigned case, once corrective action has been taken on site, the manager can mark remediation evidence as submitted, moving the case to EVIDENCE SUBMITTED and awaiting DGMS verification.

12. Field Inspector Walkthrough

12.1 Assignments

The inspector's home page lists every case currently assigned to them (ASSIGNED or INSPECTION/REMEDIATION status) plus a history of past assignments and their outcomes.

12.2 Completing a Checklist Inspection

Opening an assignment presents a four-item pass/fail safety checklist:

- Ventilation / gas monitoring functioning
- Fire-fighting / emergency equipment available
- Workers wearing required PPE
- Safety signage and barricading in place

12.3 GPS & Photo Capture

A "Capture GPS location" button uses the device's real Geolocation API to record latitude/longitude at the moment of inspection (with a clear error message if permission is denied). A photo can be attached via the device camera or file picker, plus a free-text notes field.

12.4 Offline Mode & Sync

A "Simulate offline" toggle demonstrates the field-connectivity story that's central to this product: with it on, submitting the form saves the inspection into the browser's local storage instead of sending it to the server, and shows "Pending sync: N." Turning the toggle off and pressing "Reconnect & sync" uploads every queued inspection in one batch. Each queued record carries a client-generated ID, so re-syncing the same batch twice (e.g. after a dropped connection mid-sync) never creates duplicate records - the server recognizes and ignores anything it has already received.

12.5 Filing a Standalone Field Report

The "File a report" link opens the same checklist/GPS/photo/offline pipeline described above, but without a pre-assigned case - the inspector instead picks any mine from a dropdown and a report type (Compliance Observation, Safety Incident, Environmental Observation, or Operational Exception). This is deliberately the same underlying pipeline as an assigned inspection, not a second, separate incident-reporting system - the only difference is that there is no case to attach it to yet. These reports surface to DGMS Officers on the Field Reports page (section 10.6) for triage.

13. AI & Document Intelligence

13.1 The Extraction Pipeline

When a Mine Manager uploads a document with pasted text, a background task sends that text to a locally-running Ollama instance (the Qwen 2.5 model), with an explicit instruction to extract only what is actually present: document type, permit number, issue date, expiry date, and issuing authority, returned as strict JSON. The result is defensively validated afterward - every expected field is forced to exist with the correct type regardless of what the model actually returned, and any unrecognised document type falls back to "Other" rather than failing outright.

If the extracted expiry date has already passed, the system automatically raises a DOCUMENT-sourced compliance finding against the matching KoylaNiti Control (e.g. an expired Environmental Clearance raises CMR-STAT-001) and recalculates the mine's score.

13.2 Case Brief Generation

When a case is assigned, a background task asks the same local model to write a short, neutral 2-3 sentence summary of the case's findings for the reviewing officer, with an explicit instruction never to state anything not actually present in the findings. This appears as the "AI case brief" on the case detail page.

13.3 Failure Handling

Both AI calls are wrapped so that an unreachable or misbehaving model never breaks the surrounding workflow: document extraction is simply marked FAILED (logged to the audit trail) rather than crashing the upload, and case-brief generation falls back to a plain, deterministic summary sentence built from the findings themselves. In neither case does the AI's unavailability block a case from being assigned, verified, or closed - the AI touches are additive explanation, never a dependency of the workflow itself.

14. Satellite Monitoring

The satellite comparison shown on a mine's detail page is intentionally mocked rather than a live imagery pipeline, in line with the product's own guidance to demonstrate the workflow rather than attempt a production-grade segmentation model in a hackathon prototype. A small fixed dataset provides before/after image pairs, dates, an NDVI (vegetation index) loss percentage, and an affected-area outline for two of the seeded mines. Each image pair is clearly watermarked "SIMULATED DEMO IMAGERY - not real satellite data" so it can never be mistaken for genuine remote-sensing output.

Where a mine has a satellite finding, the same underlying record also appears as an ENVIRONMENTAL-domain compliance finding, contributing to that mine's score exactly like any other finding - the mock data feeds the same fusion pipeline a real satellite pipeline would.

15. Notifications & Escalation Logic

Alerts are in-app only in this build - no email, SMS, or WhatsApp integration is claimed or implemented. Two events raise a notification automatically:

- **Near/overdue SLA on assignment:** if a case is assigned with a due date that has already passed, or falls within 48 hours, every Ministry Admin plus every DGMS Officer whose jurisdiction covers that mine's state receives an alert naming the case and its due date.
- **A CRITICAL finding is detected:** when the document-extraction pipeline raises a new finding whose rule severity is CRITICAL, the same oversight audience is notified immediately, naming the mine and the finding.

Every account also sees an unread-count badge next to "Alerts" in the header, and can mark individual alerts as read from the Alerts page.

16. Contractor Management

Contractor linkage is deliberately kept lightweight - a governance relationship attached directly to a compliance finding, not a separate contractor-management system. A finding may optionally record which contractor's work it relates to (e.g. a ventilation-monitoring contractor linked to a failed ventilation checklist finding). The contractor's profile page aggregates every finding linked to them across mines, counts their currently-open cases, and links straight through to each underlying finding and case.

17. Audit Trail

Every sensitive or state-changing action in the system writes a permanent, timestamped entry: case creation (automatic, when a mine turns RED), every case status transition (who did it, from which status to which), document extraction completing or failing, a mine's score being recomputed, and a standalone field report being reviewed. The fullest view of this is the Audit Timeline node on a case's Evidence Graph (section 8), which lists every event on that case in chronological order.

The audit log is append-only and hash-linked: every event's hash is SHA-256 of its own content plus the previous event's hash, computed at the moment it's written. No route in the application ever updates or deletes an audit row. A dedicated GET `/api/v1/audit/verify` endpoint (surfaced at `/gov/audit`, and linked directly from every case's Evidence Graph) walks the entire chain and recomputes every hash, so altering or removing so much as one historical row - by any means, not just through the application - is detectable and gets pinpointed to the exact event. This has been verified directly: editing an audit row's content in the database itself, bypassing the API entirely, is correctly caught and localized by the verification endpoint.

One honest limit: this detects tampering after the fact, it does not prevent a privileged database user from editing rows directly - there is no database-level trigger enforcing immutability, only the application's own routes (which never expose an update/delete path) and the chain-verification check. Each event's hash also excludes its own timestamp deliberately, since the timestamp is assigned by the database at insert time and is not known until after the row exists - the hash covers everything that is known and fixed at write time.

18. Security Model

- Passwords are hashed (bcrypt), never stored or logged in plain text.
- Sessions use a signed JWT (HS256), issued on login and carried as an httpOnly cookie between the browser and the web app, and as a Bearer token between the web app's server and the backend API - the raw token is never exposed to client-side JavaScript.
- Every API endpoint independently re-derives the caller's role and scope from the token against the database - the frontend's own role-based routing is a convenience, not the actual security boundary.
- Submission endpoints are rate-limited to guard against abuse.
- Evidence photos and documents are served only through access-controlled endpoints (or a same-origin authenticated proxy, for content that needs to render as a plain image tag), never as a public URL.
- The audit log is append-only and hash-linked (SHA-256 chain) - see section 17 for exactly what this guarantees and what it doesn't.
- The public complaint form (section 20) is rate-limited and honeypot-protected, and captures no identifying information at all.

19. Technology Stack

Layer	Technology
Frontend	Next.js 16 (App Router, Server Components + Server Actions), TypeScript, Tailwind CSS, Leaflet
Backend	FastAPI, SQLAlchemy 2.0, PostgreSQL, JWT authentication
AI	Local Ollama (Qwen 2.5) for document field extraction and case-brief generation
Mapping	Leaflet with OpenStreetMap tiles
Deployment	Vercel (frontend), Render (backend), Neon (managed Postgres)

20. Public Complaints (Anonymous Reporting)

Anyone - a worker, a nearby resident, anyone at all - can raise a concern about a specific mine without logging in, at `/complaint/new` (linked from the login page as "Report a concern anonymously"). This is a deliberately simplified anonymous-only model: no name, contact detail, or any other identifying information is ever asked for or recorded - there is no separate identity database, no encryption step, and no recovery-phrase system to manage, because there is no identity to protect in the first place.

20.1 Submitting a Complaint

The submitter picks the mine, a category (Safety, Environmental, Labour/Worker, Corruption/Malpractice, or Other), writes a description, and may optionally attach a photo. The form carries a hidden honeypot field real visitors never see or fill; a filled honeypot is rejected outright, and the endpoint is rate-limited (10 submissions per hour) against abuse. On success, the submitter is shown a generated complaint ID (e.g. COMP-1003) with an explicit warning that it is the *only* way to check the complaint's status later.

20.2 Checking Status

At `/complaint/status`, entering a complaint ID returns only its status and submission date - never the description, photo, or any officer's review notes. This is a deliberately minimal response: the ID is not treated as a secret credential, so nothing sensitive is tied to it being guessed or shared.

20.3 AI Summary

Every complaint gets a neutral, plain-language summary generated in the background by the same local Ollama pipeline used for document extraction and case briefs (section 13) - the third use of that one shared pipeline, not a separate system. The prompt explicitly instructs the model never to invent a fact not present in the complainant's own text, and never to pass judgement on credibility - only to restate what was said. If the model is unreachable, the summary falls back to a plain truncation of the original text rather than blocking submission.

20.4 Officer Triage and Escalation

DGMS Officers and the Ministry Admin see incoming complaints on a jurisdiction-scoped triage queue (section 10.8) - a Chhattisgarh officer never sees a Telangana complaint, and vice versa. Reviewing a complaint's detail page shows the AI summary, the full text, any photo, and two actions: **Dismiss** (with notes, if it doesn't warrant action) or **Escalate to case** (with notes). Escalating creates a brand-new Case at DETECTED, exactly as if the score engine had opened it automatically - from that point on it is indistinguishable from any other case and moves through the ordinary assign -> verify -> resolve lifecycle (section 7) with no special-casing. The resulting case's detail page shows a small "Originated from public complaint" banner linking back. A complaint never touches a mine's compliance score directly - only a verified, officer-driven case can do that, the same as every other pathway into the system.

21. End-to-End Demo Walkthrough

A single worked example tying every section above into one continuous flow, using the seeded "hero" mine (Korba West Opencast Mine, MINE-1003, Chhattisgarh):

- 1 Log in as **dgms_east**. The map shows Korba West as a RED pin, the only mine in this officer's jurisdiction.
- 2 Open the mine. The score breakdown shows 55/100, RED. The risk panel explains why: an expired Environmental Clearance, an overdue safety inspection, a failed ventilation check, and a satellite vegetation-loss anomaly - all CRITICAL or HIGH.
- 3 Open the linked case. It sits at TRIAGED, unassigned. The Evidence Graph shows the KoylaNiti Control behind each finding (including CMR 2017 Reg. 106(2)/(3) for the overdue-inspection control), the expired document, an empty Field Observation node (no inspection yet), and "Not verified yet"/"Not closed yet" for the remaining nodes.
- 4 Assign the case to Field Inspector Ramesh Patil, with a due date and an escalation target. The case moves to ASSIGNED, and (if the due date is close) an oversight alert fires.
- 5 Log in as **inspector1**. The assignment now appears on the inspector's home page.
- 6 Open it, complete the four-item checklist, capture GPS and a photo, and submit - optionally toggling "Simulate offline" first to demonstrate the save-locally / sync-later flow.
- 7 The case automatically advances to EVIDENCE SUBMITTED (shown as VERIFICATION PENDING).
- 8 Log back in as **dgms_east**, reopen the case. The Field Observation node now shows the submitted inspection, GPS coordinates, and photo. Click Verify - the linked findings become RESOLVED and the case moves to VERIFIED.
- 9 Enter a closure reason and close the case. The system rejects this step if any linked finding is still open - here it succeeds, moving the case to CLOSED and recording the Closure Evidence and Authorized Verification nodes.
- 10 Return to the mine detail page - the score has recalculated, reflecting the now-resolved findings.

This is the entire pitch in one walkthrough: documents, field inspections, and satellite signals all feed one honest, explainable score per mine, and every detected problem is tracked from automatic detection through to a verified, audited, and closed resolution - not left as a map, a chatbot answer, or a report nobody is required to act on.

A second, equally real entry point feeds the exact same pipeline from the other direction: an anonymous public complaint (section 20), once an officer escalates it, becomes a case at DETECTED and walks through steps 4 onward of this same walkthrough - assignment, field verification, and closure - completely unmodified. The audit chain behind either path can be independently verified at /gov/audit (section 17).